How TTM-r2 makes a forecast

IBM · 0.8M parameters · MLP mixer, no attention · zero-shot

1. One series, 512 steps

TTM reads exactly the last 512 steps of one series. A ? is a value not known yet.

past *future*

Target temperature

A shorter history is padded and marked as unseen.

Covariates are not used zero-shot; each metal is forecast on its own.

2. Scale, then 8 patches

The context is scaled by its own mean μ and standard deviation σ and cut into 8 patches.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

512 steps = 8 patches of 64

linear layer

9 tokens of 192 numbers

The orange token in front marks the sampling frequency.

3. Mixers instead of attention

Small MLPs mix the tokens along time, then across features. Patches are merged and split over 3 levels.

patches →

features

Time mixing: one MLP across the patches, for each feature

Feature mixing: one MLP across the 192 numbers of each patch

Gate: a softmax weighting of features, not self-attention

4. A point forecast

A small mixer decoder and a linear head give a **point forecast** for 48 or 96 steps; the first H are kept.

point forecast

IBM ships one branch per context and horizon, such as 512-48; the library picks one that covers H. No quantiles.

Tiny and attention-free: TTM forecasts in milliseconds on a CPU, but gives point forecasts only.

2 mixer layers per level · 3 levels · width 192 · decoder width 128 · patches of 64 · 512-step context

In the notebooks: 512-step branches picked per horizon · last 512 steps · each metal on its own · no covariates · no intervals